LIBBY — PATIENT/CAREGIVER TRACK

osteosarcoma-mets-dll3-h7r2

Plain-language summary for patient and caregiver

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What this page is

You (or your clinician) asked Libby to look at one specific question: **given metastatic osteosarcoma where the cancer is showing signs that DLL3 — a protein that some new immunotherapies target — might be present, what treatments would target DLL3?**Your input was that you'd consider high-risk options if the upside is meaningful.

The DLL3 information you gave us was at the **RNA level**— that tells us the gene is being read, but it doesn't tell us whether the DLL3 protein is actually sitting on the surface of the cancer cells (which is what those new drugs need to grab onto). So the recommendations come in two parts: **a first step everyone agreed on**(a confirmatory test), and then **one treatment option**that is only on the table if that test comes back positive.

This page is scoped to **drugs that target DLL3**— that's the molecular feature the question was about. Other treatments may exist for relapsed osteosarcoma in general, but they don't target DLL3, so they're outside the scope of this page. Your oncologist will know which standard options apply to your situation; that's a separate conversation from this report.

The first step everyone agreed on

Get a DLL3 protein test on your tumor — specifically an IHC stain using an antibody called SP347. Your oncologist can order this on stored tissue from a previous biopsy, or fresh tissue if needed. It's non-toxic, takes 1–3 weeks, and costs almost nothing compared to a treatment cycle. **All five reviewers ranked this as the first action — without dissent.**

The test result decides whether the treatment option below is on the table.

The option the board considered

Option 1 — Tarlatamab via clinical trial NCT06788938

This option is only available if the DLL3 IHC test comes back positive. If the test is negative or below threshold, this option is off the table, and this case has no further treatment recommendations within scope. Standard 2L+ care for relapsed osteosarcoma is the next conversation to have with your oncologist, separate from this page.

The board recommended this as the lead option *when the test is positive*.

What it is.A "bispecific T-cell engager" given as an infusion every two weeks. One end of the molecule grips DLL3 on the cancer cell, the other end recruits an immune T cell to attack it. Currently approved for small-cell lung cancer. The trial accepts patients with **any**DLL3-positive tumor, including osteosarcoma, **as long as the IHC test confirms DLL3 is on the cell surface at sufficient levels**.

What the upside might look like. In small-cell lung cancer, where this drug is approved, tarlatamab nearly halved the risk of death compared to chemotherapy in a recent randomized trial. About 40 of every 100 patients had measurable tumor shrinkage. **Whether this translates to osteosarcoma is exactly the question the trial would answer — and you'd be helping to answer it.**

The main risks.A reaction called "cytokine release syndrome" happens in about half of patients (mostly mild; about 1 in 100 are severe). Neurologic side effects in roughly 10%. The first cycle requires being in hospital for monitoring.

Where the board agreed and where they didn't.

- *Risktaker, advocate, conservative, consensus reviewer:*endorse. The
- *Critic:*still dissents — even with a positive IHC, no osteosarcoma

If the IHC is negative,this option closes entirely. There is no "backup option" within scope of this page — Libby was asked specifically about DLL3-targeted therapy. Your oncologist can talk you through standard 2L+ options for relapsed osteosarcoma (which exist and are well-established); that conversation is separate from this page.

What the board could not figure out

Several things weren't in the inputs:

- **Was DLL3 measured at the protein level (IHC) or only RNA?**Resolved by
- **What treatment have you already had?**The recommendations assume
- **What's your performance status (how active are you day-to-day)?**Most
- **Where do you live, and what's your access to academic cancer centers?**

Questions to ask your oncologist

1. **Can we order DLL3 IHC (SP347 antibody) on my tumor?**That's the first
2. **If the IHC comes back positive, what's the path to NCT06788938?**
3. **If the IHC comes back negative, what 2L+ options do you usually
4. **If we pursue tarlatamab, what's the realistic timeline from IHC
5. **What's your experience with the side effects (CRS, neurologic
6. **Have I already had any of the standard 2L+ osteosarcoma options? What
7. **Is there anything in my labs or scans that would change which trial
8. **If the tarlatamab trial isn't reachable (slot, location, eligibility),

Where to read more

- Clinician summary— the technical version of this page.
- Trial table— the trials Libby reviewed.
- Evidence list— the clinical and lab studies cited.
- Tumor-board transcript— what each reviewer actually said.
- Recommendations detail— full structured tables.

Sources

The recommendations cited the following published reports and clinical-trial records:

- Ahn et al. *N Engl J Med*2023 — tarlatamab DeLLphi-301 in SCLC (PMID 37861218)
- Mountzios et al. *N Engl J Med*2025 — tarlatamab DeLLphi-304 in SCLC
- Yao et al. *Oncologist*2022 — DLL3 as a therapeutic target review
- Wang et al. *J Exp Clin Cancer Res*2018 — Notch3/DLL3 axis in osteosarcoma
- Trial registry NCT06788938 — UCCC-01 / UCLA L-10 tarlatamab basket